

SAN MIGUEL DE TUCUMAN, Argentina

WMO: 871210

Lat: 26.8372S

Lon: 65.1083W

Elev: 1476

StdP: 13.93

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.0	40.9	26.5	21.6	56.2	29.5	24.9	53.2	17.8	61.3	15.5	58.6	6.6	320	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.3	98.1	74.4	95.1	74.2	92.8	73.7	79.6	90.9	78.2	89.3	77.0	87.9	10.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.7	147.2	87.4	75.2	139.6	85.6	73.7	132.5	83.6	44.7	91.4	43.2	89.3	41.7	88.0	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.9	15.6	13.1	DB	33.2	104.7	3.0	2.7	31.0	106.6	29.2	108.1	27.6	109.6	25.4	111.6
			WB	31.2	82.8	3.2	1.8	28.9	84.1	27.1	85.2	25.3	86.2	23.0	87.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.7	78.5	76.6	73.5	68.3	62.1	56.9	56.2	61.0	66.3	71.9	75.5	77.9	(6)
(7)		DBStd	10.03	5.21	5.59	5.49	6.07	6.00	5.19	6.38	7.57	7.93	7.27	6.50	5.75	(7)
(8)		HDD50	33	0	0	0	0	2	8	17	5	1	0	0	0	(8)
(9)		HDD65	976	0	1	7	36	130	246	282	171	77	21	4	1	(9)
(10)		CDD50	6847	883	745	728	548	376	215	209	346	490	679	764	864	(10)
(11)		CDD65	2314	418	326	269	134	39	3	8	47	116	235	319	400	(11)
(12)		CDH74	21428	4046	2792	1873	772	195	20	98	606	1349	2453	3226	3998	(12)
(13)		CDH80	9270	1837	1150	645	193	20	1	18	217	568	1162	1565	1894	(13)
(14)	Wind	WSAvg	6.5	6.9	6.6	6.0	5.7	5.4	5.6	6.2	6.7	7.2	7.2	7.2	7.1	(14)
(15)	Precipitation	PrecAvg	40.1	8.8	7.2	6.6	2.6	1.1	0.5	0.4	0.4	0.6	2.6	4.1	6.5	(15)
(16)		PrecMax	59.2	17.8	15.3	15.4	6.0	3.9	1.5	1.2	1.5	2.9	6.4	7.7	14.8	(16)
(17)		PrecMin	23.7	3.8	1.6	1.6	0.2	0.0	0.0	0.0	0.0	0.0	0.4	1.0	0.6	(17)
(18)		PrecStd	8.7	3.9	3.1	3.3	1.5	0.9	0.3	0.3	0.5	0.6	1.6	2.0	3.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.4	97.2	93.2	88.2	82.5	76.7	82.1	93.1	96.7	100.7	100.8	101.9	(19)
(20)			MCWB	77.3	78.2	77.2	74.5	69.6	65.0	64.9	68.2	70.0	72.8	73.6	75.4	(20)
(21)		2%	DB	95.2	93.3	89.6	85.1	78.9	72.7	76.2	86.0	91.3	95.3	96.6	96.7	(21)
(22)			MCWB	76.9	77.4	76.1	73.2	67.7	61.6	61.4	64.6	67.0	71.2	72.2	75.1	(22)
(23)		5%	DB	92.2	89.9	86.4	82.1	75.6	69.8	72.4	80.6	86.7	91.1	92.9	93.2	(23)
(24)			MCWB	75.9	76.5	74.4	71.0	65.6	59.0	58.7	62.1	64.8	69.1	71.7	75.0	(24)
(25)		10%	DB	89.4	87.1	83.8	78.6	72.2	66.6	68.4	75.6	82.2	86.5	89.2	89.8	(25)
(26)			MCWB	75.2	75.3	73.2	69.5	63.8	57.9	56.6	59.4	62.4	67.3	70.5	73.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.7	81.1	79.7	77.0	71.5	66.7	66.0	69.3	72.6	76.4	78.6	80.7	(27)
(28)			MCDWB	93.1	91.7	89.2	85.9	79.2	73.8	79.4	88.7	89.0	93.8	91.5	93.3	(28)
(29)		2%	WB	79.4	79.3	77.6	74.5	69.2	63.8	63.2	66.2	69.4	73.7	76.2	78.4	(29)
(30)			MCDWB	90.8	89.7	86.7	82.1	76.1	70.0	73.6	83.3	86.5	89.6	89.6	91.0	(30)
(31)		5%	WB	77.8	77.7	76.0	72.6	67.4	61.6	60.7	63.5	67.0	71.5	74.3	76.8	(31)
(32)			MCDWB	88.8	87.5	84.3	79.3	73.4	67.1	69.2	78.1	81.8	85.2	87.4	89.6	(32)
(33)		10%	WB	76.3	76.2	74.2	70.8	65.5	59.5	58.1	60.8	64.5	69.5	72.4	75.2	(33)
(34)			MCDWB	86.6	85.2	81.6	76.9	71.0	64.7	66.3	73.3	79.1	82.3	84.7	86.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.3	16.6	15.4	15.7	15.7	17.4	21.2	24.2	24.6	22.1	21.0	19.7	(35)
(36)			MCDBR	22.8	21.4	20.4	20.8	20.8	22.7	26.9	31.2	31.8	30.2	27.8	25.5	(36)
(37)			MCWBR	10.3	10.1	10.2	11.0	11.5	13.0	14.4	14.9	13.9	12.1	11.6	11.1	(37)
(38)		5% WB	MCDWB	20.5	19.4	18.4	17.6	17.6	18.5	23.4	28.5	27.7	25.3	22.8	22.1	(38)
(39)			MCWBR	10.4	10.0	10.1	10.1	10.4	11.5	13.4	14.2	12.9	11.1	10.8	10.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.408	0.407	0.392	0.370	0.331	0.318	0.318	0.361	0.429	0.433	0.414	0.413	(40)	
(41)		taud	2.390	2.388	2.430	2.447	2.508	2.510	2.478	2.346	2.156	2.227	2.325	2.368	(41)	
(42)		Ebn at Noon	297	292	287	278	276	273	278	276	271	281	293	296	(42)	
(43)		Edh at Noon	41	40	36	33	28	26	28	35	47	46	43	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1997	1823	1531	1277	1062	1000	1159	1451	1740	1881	2015	1988	(44)	
(45)		RadStd	150	145	127	112	107	89	75	87	138	155	177	129	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.71	+0.67	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	+1.37	+0.81	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page